

Human Computer Interaction

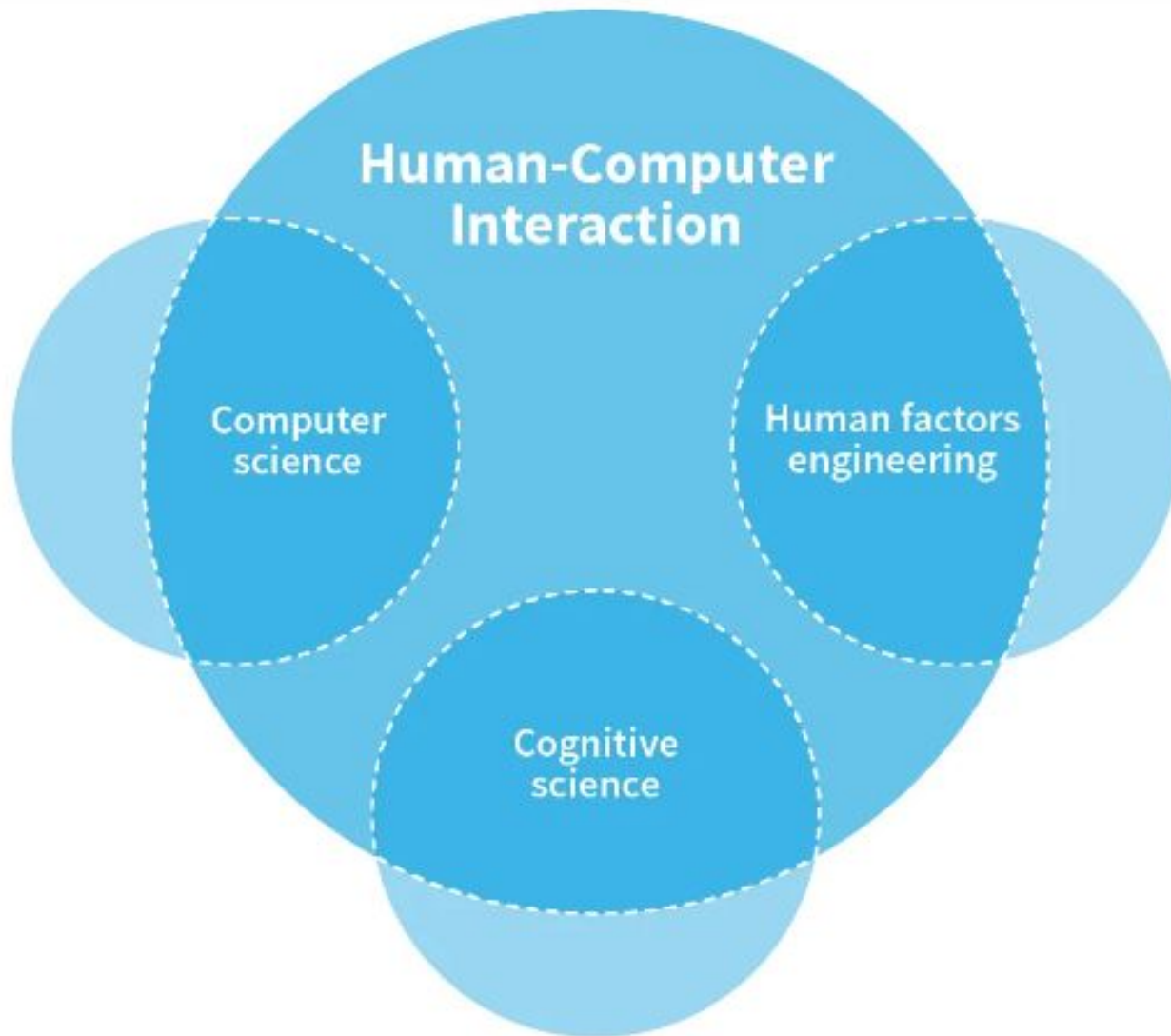
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Introduction

- Human-computer interaction (HCI) is the field of study that focuses on optimizing how users and computers interact by designing interactive computer interfaces that satisfy users' needs.
- Human-computer interaction (HCI) is a multidisciplinary field of study focusing on the design of computer technology and, in particular, the interaction between humans (the users) and computers.
- It is a multidisciplinary subject covering computer science, behavioral sciences, cognitive science, ergonomics, psychology, and design principles.
- HCI is really the place where communication between person and machine takes place, generally a display of some kind and then various other tools for sending data to and getting data from the computer.
- This interaction between person and machine is called the “loop of interaction”.
- Today, the most commonplace tools for maintaining this loop are keyboards and some sort of pointing device like a mouse. And, of course, touchscreens are everywhere these days, especially on laptops, tablets and smartphones.



Ergonomics

- Ergonomics is the science of designing and arranging objects, spaces, and systems to fit the people who use them, aiming to optimize human well-being and overall system performance.
- It's essentially about making sure that the tools, equipment, and environments people interact with are comfortable, efficient, and safe.
- This discipline considers factors such as human anatomy, physiology, psychology, and biomechanics to create products and environments that minimize discomfort and risk of injury while maximizing productivity and usability.

Ergonomics

- Ergonomics in human-computer interaction (HCI) is crucial for designing interfaces and systems that optimize user comfort, efficiency, and safety.
- By integrating ergonomic principles into HCI design, developers can create systems that enhance user satisfaction, productivity, and overall well-being.
- Additionally, considering the diversity of users and their varying needs is fundamental to creating inclusive and accessible interfaces.

Types of Ergonomics

- **Physical Ergonomics:**

- This focuses on the design of physical elements such as chairs, desks, keyboards, and mice.
- It involves ensuring proper posture, minimizing repetitive strain injuries (RSIs), and providing adjustable equipment to accommodate various body types and preferences.

- **Cognitive Ergonomics:**

- This concerns the mental processes involved in using computer systems.
- It includes factors like information organization, task complexity, and the clarity of instructions.
- Cognitive ergonomics aims to reduce cognitive load and mental fatigue by designing interfaces that are intuitive and easy to understand.

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- **Visual Ergonomics:**

- This addresses the visual aspects of computer interfaces, such as font size, contrast, and color schemes.
- Good visual ergonomics can prevent eye strain, headaches, and other visual discomforts.
- Design choices should consider readability, accessibility, and the preferences of diverse user groups.

- **Interactive Ergonomics:**

- This focuses on the usability of interactive elements like buttons, menus, and navigation controls.
- It involves designing interfaces that are responsive, predictable, and conducive to efficient interaction.
- Feedback mechanisms, error prevention, and user customization options are important considerations.

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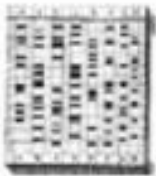
- **Environmental Ergonomics:**

- This considers the broader physical environment in which computer systems are used.
- Factors like lighting, noise levels, temperature, and spatial layout can affect user comfort and productivity.
- Designing workspaces that are conducive to concentration and well-being is essential.

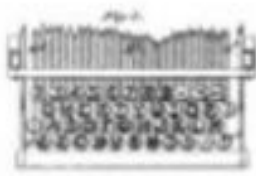
Evolution of HCI



Evolution of HCI



Punch Cards for
Informatics
1832



QWERTY
Keyboard
1872



Electromechanical
Computer (Z3)
1941



Electronic Computer
(ENIAC)
1943



Paper Tape Reader
(Harvard Mark I)
1944



Mainframe Computers
(IBM SSEC)
1948



Trackball
1952



Joystick
1967



Microcomputers
(IBM Mark-8)
1974



Portable Computer
(IBM 5100)
1975



Commercial Use of
Window-Based GUI
(Xerox Star)
1981



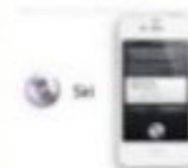
Commercial Use
of Mouse
(Apple Lisa)
1983



Commercial Use
of Mobile
Computing
(PalmPilot)
1996



Touch + Camera -
based Mobile
Computing
(iPhone 2G)
2007



Voice on Mobile
(Siri)
2011



Voice on Connected /
Ambient Devices
(Amazon Echo)
2014

Evolution of HCI

- The evolution of Human-Computer Interaction (HCI) has been shaped by technological advancements, changes in user needs and behaviors, and shifts in design paradigms.
- Throughout its evolution, HCI has become increasingly interdisciplinary, drawing from fields such as psychology, sociology, anthropology, and design.
- The focus has shifted from merely making computers usable to creating meaningful and enjoyable interactions that enhance human capabilities and experiences.

Early Computers

- When we think of the early days of computing, we might think of giant, room-sized machines, filled with vacuum tubes, using punch cards to maintain the loop of interaction. These were built in 1937 by Atanasoff-Berry Computer (ABC), and would eventually become the Electronic Numerical Integrator and Computer (ENIAC) in 1946.
- ENIAC weighed 30 tons and had more than 18,000 vacuum tubes. It could only perform one task at a time and has no OS.
- Transistor-based computers were smaller, like the 1951 UNIVAC 1 and, in 1953, the IBM 600 and 700 series.
- Between that time and 1962, over 100 computer programming languages were developed, and computers had operating systems, memory, storage and access to printers.
- During this period interactions with computers were primarily through direct manipulation of physical controls or through specialized input devices. Here are some methods:
- **Toggle Switches and Panels:** Early computers, like the ENIAC and Colossus, were programmed by physically flipping switches or setting jumper wires on panels. Each switch or wire represented a specific binary instruction or data input.
- **Punched Cards:** In the mid-20th century, punched cards became a common method for inputting programs and data into computers. Users would punch holes into cards according to a specific encoding scheme, with each card representing a line of code or a data entry. These cards were then fed into card readers attached to the computer.

Early HCI (1960-1980s)

- In 1963, the integrated circuit changed things again, giving birth to computers we might think of as modern.
- During this period, computers were primarily used by experts and researchers. HCI focused on designing interfaces for specialized tasks and user groups.
- Interaction was mainly through command-line interfaces, which required users to input text commands through keyboards.
- MS-DOS was developed by Microsoft as a command-line operating system for IBM-compatible personal computers. Its development began in 1981, and the first version, MS-DOS 1.0, was released in 1981 for the IBM PC.
- MS-DOS provided a command-line interface where users interacted with the system by typing commands. It supported file management, program execution, and hardware control through commands like COPY, DIR, and FORMAT.
- It was a single-tasking operating system, meaning it could only run one program at a time.

Graphical User Interfaces (1980s)

- In 1984, Apple has launched Macintosh icon-based interaction system which is a GUI based system.
- The introduction of graphical user interfaces (GUIs) revolutionized HCI.
- When Apple launched the Macintosh in 1984, it was a game changer. People no longer had to be experts to use computers, so interaction became much easier.
- This was the time of the keyboard, mouse and icon-based interface.
- At that time, HCI gave birth to cognitive models in how to design UIs, using concepts like “folders” and “a desktop”, because people mainly used computers to complete tasks they had been performing pre-computer.
- Microsoft’s Windows operating system then launched in the 1990s.
- GUIs, popularized by systems like the Xerox Star and later the Apple Macintosh, allowed users to interact with computers using visual elements like icons, windows, and menus.

Internet and Web Interfaces (1990s):

- Then the world wide web made the internet accessible to all.
- One important feature of the new connected web was that people were interacting with computers in order to interact with one another.
- Computers became tools for communication as well as tasks, and HCI design ideas shifted from how to accomplish tasks to encouraging interaction.
- The proliferation of the internet brought new challenges and opportunities for HCI.
- Web interfaces emerged, allowing users to interact with information and services over the internet.
- Design considerations expanded to include issues like information architecture, navigation, and accessibility.

Ubiquitous Computing (2000s):

- Ubiquitous computing, also known as pervasive computing or ambient intelligence It is designed to be unobtrusive and transparent, seamlessly integrating into the environment without drawing attention to itself.
- Devices are often small, discreet, and hidden from view, allowing users to focus on their tasks without being distracted by the technology.
- It refers to the concept of embedding computing technology into everyday objects and environments, making them capable of sensing, processing, and responding to user needs without requiring explicit user interaction.
- The goal is to create seamless and unobtrusive interactions between users and computing devices, enhancing the functionality and intelligence of the surrounding environment.
- The rise of mobile devices and other embedded technologies led to the concept of ubiquitous computing.
- HCI expanded beyond traditional desktop interfaces to include interactions with a variety of devices and environments.

key aspects of ubiquitous computing:

- **Embedded Devices:** Ubiquitous computing involves embedding computing capabilities into a wide range of objects and environments, including household appliances, wearable devices, vehicles, infrastructure components, and even everyday items like furniture and clothing. These devices typically include sensors for gathering data, processors for local computation, and communication modules for connecting to networks.
- **Context Awareness:** Ubiquitous computing systems are designed to be context-aware, meaning they can sense and adapt to the changing context of their environment and users. They use sensor data, environmental cues, and user behavior patterns to infer the user's intentions, preferences, and needs, allowing them to provide personalized and contextually relevant services.
- **Ubiquitous Connectivity:** Ubiquitous computing relies on pervasive connectivity technologies such as wireless networks (e.g., Wi-Fi, Bluetooth, RFID), cellular networks, and the Internet of Things (IoT) protocols. These networks enable seamless communication and data exchange between interconnected devices, allowing them to collaborate and coordinate their actions.

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- **Intelligent Interfaces:** Ubiquitous computing systems often feature natural and intuitive user interfaces that enable effortless interaction with the technology. These interfaces may include voice commands, gestures, touchscreens, and augmented reality displays, allowing users to interact with the computing environment in a manner that feels natural and intuitive.
- **Adaptive Services:** Ubiquitous computing systems can dynamically adapt their behavior and services based on the user's context, preferences, and historical interactions. They use machine learning algorithms and AI techniques to analyze sensor data, learn from user feedback, and continuously improve their performance over time.
- **Privacy and Security:** As ubiquitous computing involves collecting and processing large amounts of personal data, privacy and security are major concerns. Designing systems with built-in privacy safeguards, encryption mechanisms, access controls, and transparent data handling practices is essential to ensuring user trust and compliance with privacy regulations.

Key characteristics of ubiquitous computing

- **Invisibility:** Ubiquitous computing technology is designed to be unobtrusive and transparent, seamlessly integrating into the environment without drawing attention to itself. Devices are often small, discreet, and hidden from view, allowing users to focus on their tasks without being distracted by the technology.
- **Context Sensitivity:** Ubiquitous computing systems are context-aware, meaning they can sense and respond to changes in their environment, user behavior, and social context. They use sensors, actuators, and artificial intelligence algorithms to adapt their behavior based on the current context, providing personalized and contextually relevant services.
- **Interconnectedness:** Ubiquitous computing devices are interconnected through networks, enabling them to communicate and collaborate with each other. This interconnectedness allows for seamless data sharing, collaboration, and coordination across devices, creating a unified computing environment.
- **Autonomy:** Ubiquitous computing devices are often autonomous, capable of operating independently without direct human intervention. They can perform tasks automatically based on predefined rules, user preferences, or environmental conditions, enhancing convenience and efficiency.
- **Natural Interfaces:** Ubiquitous computing systems often feature natural and intuitive user interfaces, allowing users to interact with the technology using familiar gestures, speech commands, or physical actions. These interfaces are designed to be user-friendly and accessible to people of all ages and abilities.

Development in Ubiquitous Computing

- Touch Screen:
 - The launch of the iPhone in 2007 popularized touch in the mobile phones as a new way of thinking about HCI.
 - Touchscreens started becoming more common for personal devices, and started showing up on computers, laptops and tablet computers. With touch technology, interactive digital signage became a real possibility.
 - Early versions of touchscreens were expensive, single-touch, professional-grade displays. Then the multi-touch screen came out, and surfaces that were static could suddenly be interacted with just like the iPhone. As more display manufacturers jumped into touch technology, prices came down and even static displays could be made interactive using overlays.
- Gesture based Interaction
- Voice Recognition:
 - All of the previous computer interface technologies we've covered have been physical transducers for Muscle-Computer Interfaces (muCIs). They are physical objects that use human muscles to do most of the interacting.
 - Today, we're seeing technologies such as Voice-Activated Interfaces (VUIs) that allow interaction with a screen without touching it at all.
 - You just speak some commands or questions to it, and it shows you what you want. It's very similar to smart speakers in homes, like Amazon's Alexa, and computer systems that already allow this type of interaction, like Microsoft's Cortana.
 - It aims to satisfy an audience used to hands-free commands with smartphone apps like Google Assistant, Siri and Bixby.

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- Facial recognition
 - It is the newest kid on the block. Screens can have cameras embedded or attached to them, and the sign “looks” at a person in front of it to recognize their facial expression.
 - Even more advanced face recognition technology is making it possible for that camera to see more – the person’s height, gender, age, ethnicity, whether they are alone or with their family, and so on.
 - It then decides what content might be relevant to that viewer, based on given sets of parameters.

Future of Human-Signage Interaction

- So, we've gone from punch cards and keyboards to simply talking to our devices, or interacting with a device by using another device. Soon, just standing in front of a device will be a type of interaction of its own.
- As AI programs and machine learning get more and more sophisticated, we'll start seeing content specifically tailored to us as unique individuals, simply by walking up to the screen and looking at it. Soon it will even be able to recognize the actual individual looking at the screen, and access their entire browsing and purchasing history to tailor content to that specific person.
- This is a type of HCI that is almost completely hidden, and so seems more natural. The interaction, through RFID, NFC, Bluetooth and cameras, becomes passive, at least on the viewer's end of things. People go about their day in the world, and the screens around them serve up content and offer things to them.

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- There's been a lot of talk about gesture interfaces. The film "Minority Report" popularized that idea, and quite a bit of progress has been made in this technology (which was based on an actual prototype called g-speak by Oblong Industries). We can already see some basic form of gesture interaction on the market and at trade shows now. Expect to see a lot more in the next few years.
- We also have early versions of eye gaze tracking (EGT) software that can see where a person is looking on a screen. A certain blink pattern can make the text larger or clearer. By using EGT-only, or even hybrid systems, interactions can be much faster than they are now.
- There is even talk that, some day soon, people will begin to embed tech into their bodies. This tech will then interact with the devices around us. Such embedded tech will leverage micro-expressions and gestures, things we humans do unawares when thinking about something.

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- Right now, some companies are experimenting with tooth-click control for devices (combined with head movements), enabling paralyzed or partly paralyzed people full access to a computer interface.
- If this is successful, it will probably trickle out into the general market as well. And then there's BCI, or brain-computer interfaces. Sounds like science fiction, but this is actually being worked on in earnest.
- All this technology will probably show up in the computer and gaming markets first, but will quickly move into advertising, as well as the realm of organizational communications and digital signage.

User Experience (UX) Design (2010s):

- The focus shifted from just usability to the broader concept of user experience (UX).
- HCI practitioners began considering emotional, social, and contextual aspects of interaction.
- Design thinking, user-centered design (UCD), and iterative design methodologies gained traction.

Emerging Technologies (2010s-2020s)

- HCI continues to evolve with the emergence of new technologies such as virtual reality (VR), augmented reality (AR), and voice interfaces.

Design challenges include creating immersive and natural interfaces, addressing privacy and ethical concerns, and adapting to diverse user needs and preferences.

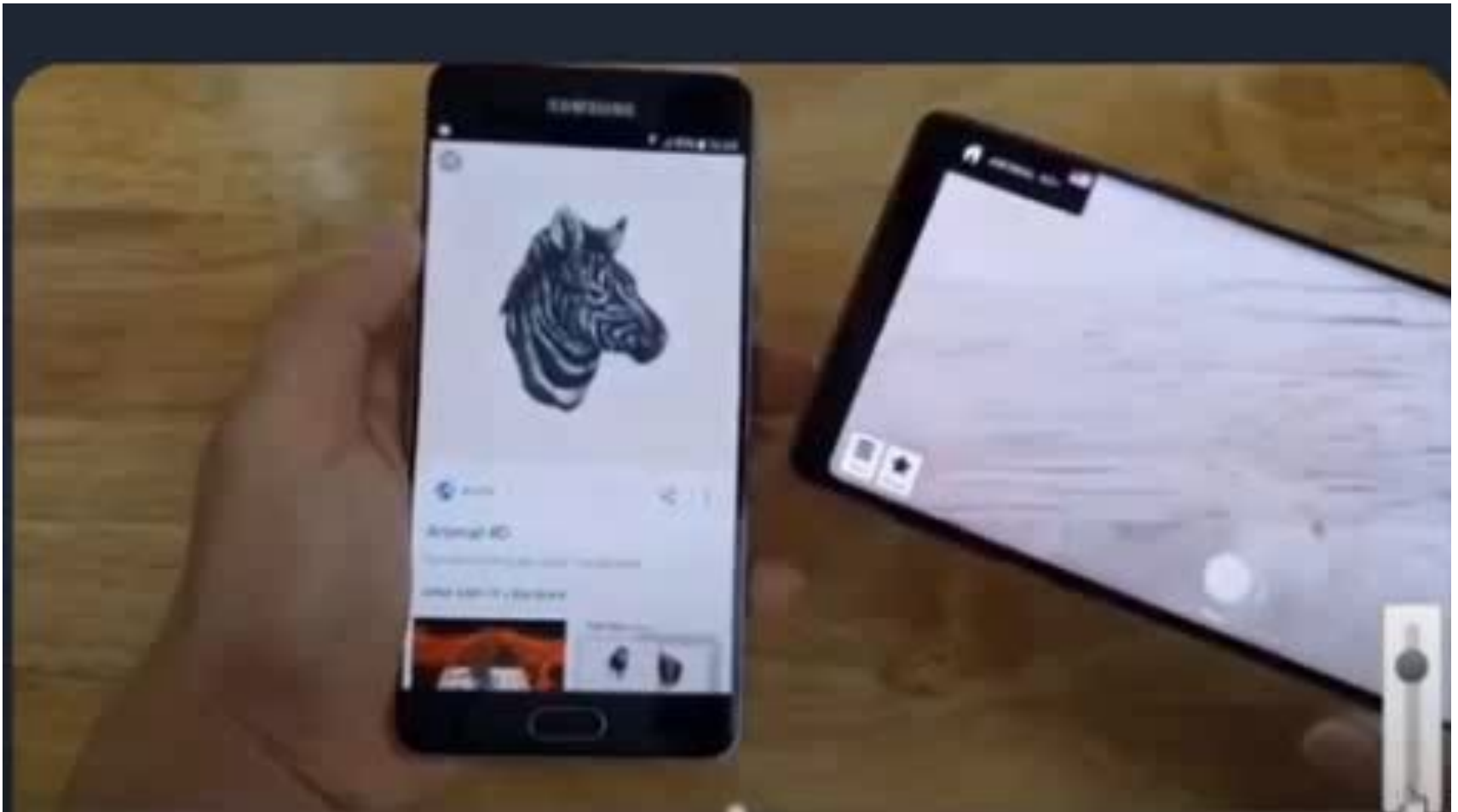
Augmented Reality

- Augmented reality is an interactive experience that enhances the real world with computer-generated perceptual information.
- Using software, apps, and hardware such as AR glasses, augmented reality overlays digital content onto real-life environments and objects.

Marker Based Augmented Reality



Marker Based Augmented Reality



Augmented Reality in Ecommerce



Virtual Reality

- VR is the most widely known of these technologies.
- It is fully immersive, which tricks your senses into thinking you're in a different environment or world apart from the real world.
- Using a head-mounted display (HMD) or headset, you'll experience a computer-generated world of imagery and sounds in which you can manipulate objects and move around using haptic controllers while tethered to a console or PC.

Looking Ahead

- Soon, everything will be interactive, but won't need to be actually touched. In fact, the very concept of digital signage might well change.
- If people have embedded tech in their bodies, then the space for communications to reach them might extend out from a physical display to a certain part of a building – the whole space, like a building lobby.
- In fact, there may be no need for a physical screen, if you think of wearable computing devices, holograms and other proposed technologies.

Looking Ahead

- As people move through the lobby, the content management system would be communicating with them, sending them messages that are specifically relevant and interesting to them as individuals, and no two people would get exactly the same content at the same time.
- And they can interact with the content they're receiving right then and there, with minimal gestures, blinks, tooth clicks, or what have you. The entire space becomes a tailored content zone for each person in it.
- This is really the ultimate hands-free and personalized experience. It's starting now with voice-activated interfaces, gesture controls and eye-gaze tracking, and will bloom into completely new ways to communicate. In 50 years, people will look back and say, "Wow, did they really used to actually touch screens with their fingers?"

Input /Output Channel

- In human-computer interaction (HCI), input and output channels refer to the means by which users interact with a computer system and receive feedback from it.

Input Channels

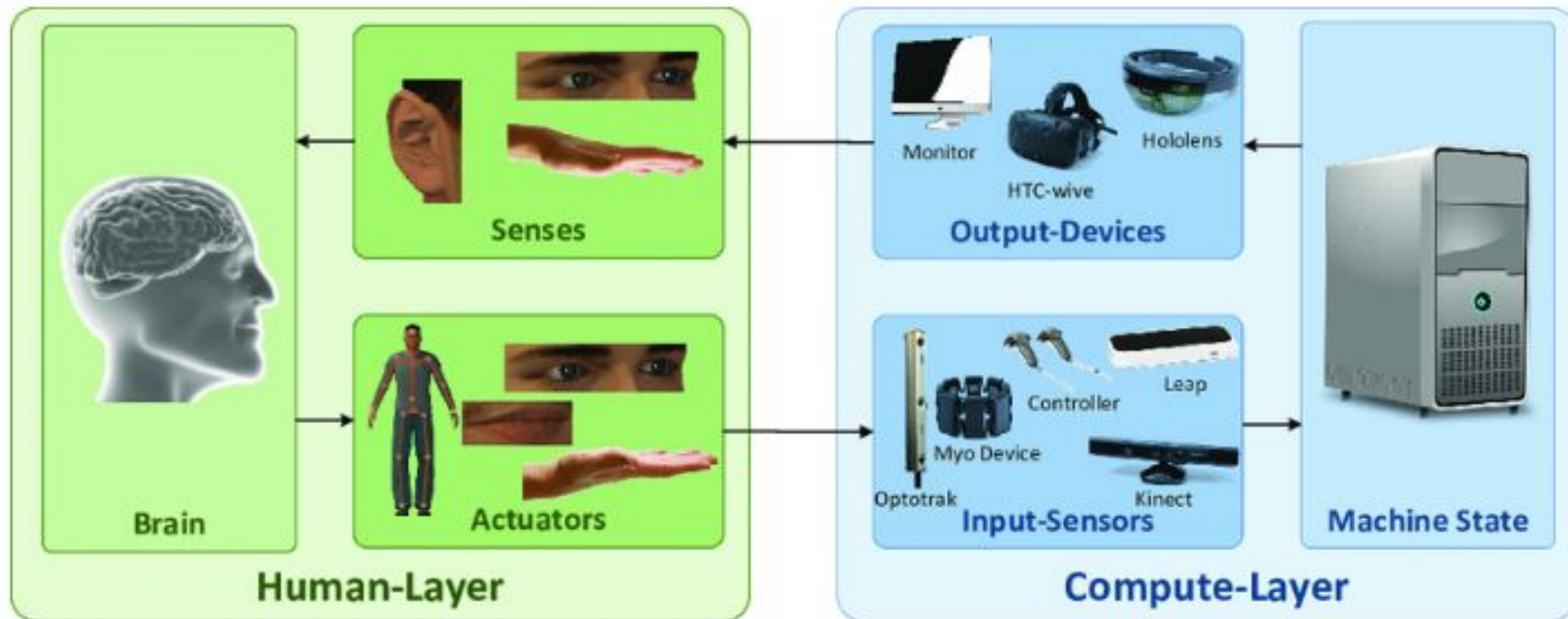
These are the mediums through which users provide information or commands to the computer system.

- Examples include:
 - Keyboards: Typing text, entering commands.
 - Mouse/Touchpad: Pointing, clicking, dragging.
 - Touchscreens: Direct touch input.
 - Voice Recognition: Speaking commands.
 - Gestures: Hand movements captured by cameras or sensors.
 - Sensors: Motion sensors, accelerometers, etc.
 - Stylus: Pen-like devices for drawing or input.

Output Channel

- These are the mediums through which the computer system provides information or feedback to the user.
- Examples include:
 - Display Screens: Monitors, LCD screens, projectors.
 - Speakers: Audio output, such as voice feedback or sound effects.
 - Haptic Feedback: Vibrations, force feedback.
 - Printers: Producing physical copies of information.
 - LEDs/Lights: Visual indicators, status lights.
 - Tactile Displays: Braille displays, tactile feedback surfaces.

Input/output Channels



Role of Human Memory in Human Computer Interaction

- Human memory plays a significant role in human-computer interaction (HCI) in several ways:
- **Recall of Interface Elements:** Users rely on their memory to recall the location and functionality of various interface elements such as menus, buttons, and icons. A well-designed interface takes into account human memory limitations, ensuring that elements are placed logically and consistently to facilitate easy recall.
- **Learning and Adaptation:** Users learn and adapt to interfaces over time through repeated interactions. This involves forming mental models of how the system works and remembering strategies for accomplishing tasks efficiently. Designers can support this process by creating interfaces that are intuitive and consistent, reducing the cognitive load required for learning.
- **Error Recognition and Recovery:** Memory aids users in recognizing errors and recovering from them. For example, when encountering an error message, users may rely on their memory of previous interactions to determine the cause of the error and how to resolve it.
- **Retrieving Information:** HCI often involves tasks that require users to retrieve information from memory, such as recalling usernames and passwords or retrieving previously accessed files. Designing interfaces that support efficient information retrieval can enhance usability and productivity.

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- **Attention and Working Memory:** Working memory, a component of human memory responsible for temporarily storing and manipulating information, plays a crucial role in HCI. Users allocate attention and working memory resources to interact with interfaces, process information, and make decisions. Designing interfaces that minimize cognitive load and support users in managing their attention effectively can improve usability.
- **Personalization and Customization:** Memory also influences users' preferences and habits, which shape their interactions with computer systems. Interfaces that support personalization and customization allow users to leverage their memory of past interactions to tailor the system to their preferences, increasing satisfaction and usability.
- Overall, human memory influences how users interact with computer systems, from learning and adaptation to error recognition and information retrieval. Designing interfaces that align with human memory capabilities can enhance usability, efficiency, and user satisfaction in HCI.

Role of Human Psychology in Human Computer Interaction

- Human psychology plays a crucial role in human-computer interaction (HCI) as it helps designers understand how users perceive, think, and behave when interacting with computer systems.
- Here are some key aspects of human psychology and their roles in HCI:
 - **Perception:** Human perception influences how users interpret and interact with interface elements. Designers consider principles of visual perception, such as Gestalt principles (proximity, similarity, closure, etc.), to organize interface elements in a way that facilitates quick and accurate comprehension. Understanding how users perceive colors, shapes, and patterns helps designers create interfaces that are visually appealing and easy to navigate.

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- **Cognitive Processes:** Cognitive psychology principles guide the design of interfaces that align with users' cognitive capabilities. This includes considerations such as attention, memory, problem-solving, and decision-making. For example, designers aim to minimize cognitive load by presenting information in a clear and concise manner, leveraging recognition rather than recall, and providing feedback to support users' mental models.
- **Emotion:** Emotions play a significant role in user experience and can influence users' perceptions, attitudes, and behaviors towards computer systems. Designers aim to evoke positive emotions through elements such as aesthetics, feedback, and usability, while minimizing negative emotions by reducing frustration and anxiety. Emotionally engaging interfaces enhance user satisfaction and promote continued use.

- **Motivation:** Understanding motivational factors helps designers create interfaces that encourage user engagement and participation. Principles from motivational psychology, such as goal-setting, feedback, and rewards, inform the design of interactive systems that motivate users to achieve desired outcomes. Gamification techniques, for example, leverage principles of intrinsic and extrinsic motivation to increase user engagement and enjoyment.
- **Social Influence:** Human-computer interaction often occurs in social contexts, where users may be influenced by social norms, peer pressure, and social identity. Designers consider social psychology principles to create interfaces that support social interaction, collaboration, and communication. Features such as social sharing, user profiles, and community forums facilitate social engagement and contribute to a sense of belonging.
- **User Diversity:** Human psychology also recognizes the diversity among users in terms of age, culture, gender, abilities, and preferences. Designing inclusive interfaces requires an understanding of how different psychological factors interact with individual differences. Designers employ techniques such as user personas, user testing, and accessibility guidelines to ensure that interfaces accommodate the needs and preferences of diverse user groups.

Pointing and Drawing

- Pointing devices allow the user to point, position and select items, either directly or by manipulating a pointer on the screen.
- Many pointing devices can also be used for free-hand drawing although the skill of drawing with a mouse is very different from using a pencil.
- The mouse is still most common for desktop computers.

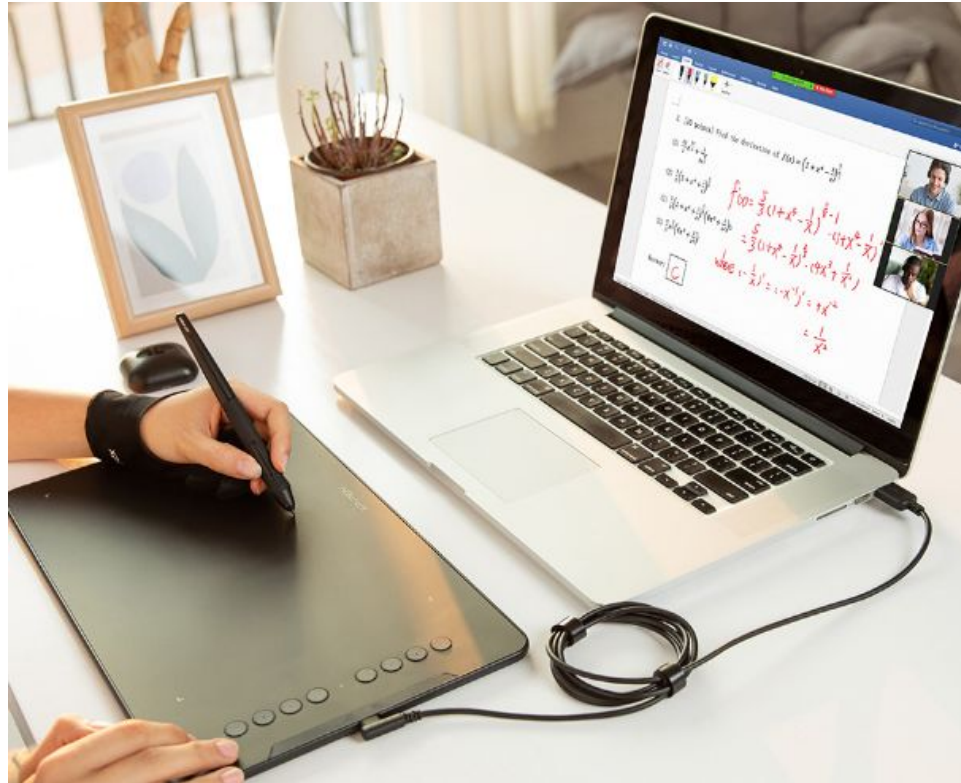
Stylus

- A **stylus** is a small pen-shaped instrument whose tip position on a computer monitor can be detected.
- It is used to draw, or make selections by tapping.
- While devices with touchscreens such as newer computers, mobile devices, and graphics tablets can usually be operated with a fingertip, a stylus provides more accurate and controllable input.
- The stylus has the same function as a mouse or touchpad as a pointing device; its use is commonly called pen computing.

Use of Stylus in track-pad and touchscreen based devices



PenTab



Virtual Reality

- VR is the most widely known of these technologies.
- It is fully immersive, which tricks your senses into thinking you're in a different environment or world apart from the real world.
- Using a head-mounted display (HMD) or headset, you'll experience a computer-generated world of imagery and sounds in which you can manipulate objects and move around using haptic controllers while tethered to a console or PC.

Types of Virtual Reality

Following three types of virtual reality technologies are available according to the experience they offer.

1. Fully Immersive VR
2. Non-Immersive VR
3. Semi-Immersive VR

Fully Immersive Virtual Reality

Fully immersive types of virtual Reality provide users with a more realistic experience. In other words, users think they are in another world. To experience a fully immersive VR, users need to wear a VR headset, motion detection sensors, and feedback devices.

Matrix world is the best example of fully immersive virtual reality. Existing technologies are not advanced like the matrix world. But they are improving with time.

Non Immersive Virtual Reality



The non-Immersive type of Virtual Reality simulates only a subset of user sense. User senses are affected by the outside world.

For example, in 3D cad tools, engineers visualize a virtual product from various aspects. But they are not immersed in it. But nowadays cad tools are also offering VR solutions to their customers.

Semi-Immersive Virtual Reality



The Semi-immersive type of VR lies in between fully immersive and non-immersive VR. In this, the users are partially immersed in the VR system. For example, in a car driving simulator, the driver sits in a cabin, and road pictures with obstacles come upon a 2D screen. Therefore, the driver is fully aware of what is happening around him.

Key Characteristics of Virtual Reality

VR system shall have the following key characteristics to make the VR experience more realistic. The higher the sense of these characteristics, the user experience will be more realistic.

Believable Virtual World

The believable virtual world is a key characteristic of Virtual Reality. The virtual world is a computer-generated responsive 3-Dimensional virtual environment where real-world rules are applied.

Therefore VR experience can become more realistic if the 3D virtual world has the following characteristics.

- Closer to the real world.
- Responsive.
- Interactive.
- Fast.
- Bug-free.

For example, designing a VR application where a human can fly, or jump a 20-meter wall will not look realistic. And the user will not be able to immerse himself in the virtual world because you are not creating a realistic VR experience.

Immersive

VR needs to interact with user senses to ensure users believe they are physically present in the real world while being in a virtual environment. User experience in VR depends on the number of human senses a VR device can control.

The best VR device can control all five human senses. But technology to control all five human senses is not available to date. Matrix movie is the best example where machines control all human senses.

Feedback

Getting feedback is a key characteristic of virtual reality. It makes the VR experience more realistic and interactive.

For example, while playing boxing in a VR environment. VR experience will be more realistic if user experiences that hit through vibration sound or any other means.

Interactive

The virtual world should be interactive and fast to make the VR experience more realistic. For example, the VR experience is more realistic when the virtual environment responds to user action naturally and with minimal lag.

Devices of Virtual Reality and 3D Interaction

- Virtual reality (VR) devices are hardware systems that immerse users in a simulated environment, creating a sense of presence and interaction with the digital world.
- These devices typically consist of:
 - VR Headset
 - Controllers
 - Sensors
 - Computing Hardware

VR Headset

- A VR headset system provides real-time viewer-centered head-tracked perspective with a large angle of view and binocular (stereo) display.
- A **virtual reality headset** (or **VR headset**) is a head-mounted device that uses 3D near-eye displays and positional tracking to provide a virtual reality environment for the user.
- VR headsets typically include a stereoscopic display (providing separate images for each eye), stereo sound, and sensors like accelerometers and gyroscopes for tracking the pose of the user's head to match the orientation of the camera present in the virtual world with the user's eye positions in the real world.
- The head-tracking changes the field of vision as a person turns their head.
- The technology may not be perfect, as there is latency if the head moves too fast. Still, it does offer an immersive experience.
- Head Mounted Displays (HMDs) like the Oculus Rift achieve these features by using small display screens that move with the viewer, close to the viewer's eyes.



VR Headset

- The basic technique of stereoscopic displays is to present offset images that are displayed separately to the left and right eye. Both of these 2D offset images are then combined in the brain to give the perception of 3D depth
- Some VR headsets also have eye-tracking sensors.
- Virtual reality headsets and viewers have also been designed for smartphones. Unlike headsets with integrated displays, these units are essentially enclosures which a smartphone can be inserted into.
- VR headsets are widely used with VR video games, but they are also used in other applications, including simulators and trainers.

Controllers and Input Devices

- VR systems often come with controllers or other input devices that allow users to interact with virtual environments.
- These can range from handheld motion controllers with buttons and triggers to gloves that track hand movements.



Sensors

- External sensors or cameras that track the position and movement of the headset and controllers in physical space, enabling accurate representation of the user's movements within the virtual world.



Computing Hardware:

- VR experiences require significant computational power to render high-resolution graphics and maintain low latency.
- This can include powerful PCs or gaming consoles that meet the minimum requirements specified by the VR headset manufacturer.

Virtual Reality Headset

A high-end VR headset consists of a display unit and earphones for audio simulation and gets input from computing devices such as X-box or PlayStation. Whereas low-cost VR devices utilize smartphone screens as display and smartphone vibrations provide haptic feedback.



- A VR headset is a visual screen or display. Headsets often include state-of-the-art sound, eye or head motion-tracking sensors or cameras.
- There are three main types of headsets: PC-Based VR Headsets: PC headsets tend to be the highest-priced devices because they offer the most immersive experiences.

Following are some of the popular virtual reality Headsets (head-mounted display) available in the market.

- [HTC VIVE](#)
- [Oculus Rift](#)
- [PlayStation VR](#)
- [Samsung Gear VR](#)
- [Google Cardboard](#)

Evaluation of HCI

Valuation of human-computer interaction (HCI) involves assessing how effectively and efficiently users can interact with a system, along with their overall satisfaction.

Key Aspects of HCI Evaluation:

- **Usability:** Focuses on how easy and intuitive it is for users to learn and operate the system to achieve their goals.
- **User Experience (UX):** Involves users' perceptions and responses to the system, encompassing emotional, subjective, and behavioral aspects.
- **Task Performance:** Measures how well users can accomplish specific tasks with the system, including metrics like accuracy, efficiency, and task completion time.
- **Effectiveness:** Refers to the ability of users to achieve their goals accurately and completely with the system.
- **Efficiency:** Relates to the speed and resources required for users to accomplish tasks.
- **Satisfaction:** Captures users' overall attitudes and satisfaction levels regarding their interaction with the system.

Methods of Evaluation for Human Computer Interaction

- There are several methods to evaluate human-computer interaction (HCI), each serving different purposes and focusing on various aspects of the interaction. Here are some common methods:
 - Usability Testing
 - Heuristic Evaluation
 - Surveys and Questionnaires
 - A/B Testing
 - Task Analysis
 - Performance Metrics:
 - Eye Tracking
- The choice of method depends on factors such as the stage of development, the goals of the evaluation, available resources, and the specific aspects of HCI being evaluated (usability, user satisfaction, efficiency, etc.).
- Often, a combination of methods is used to gain a comprehensive understanding of the user experience.

Usability Testing

- Usability testing focuses on evaluating how well users can interact with a system to achieve their goals efficiently, effectively, and satisfactorily.
- It Involves observing users as they interact with the system to identify usability issues. This can be done through thinking-aloud protocols, where users verbalize their thoughts while performing tasks.

Purpose of Usability Testing

- **Identifying Issues:** Usability testing helps uncover usability problems that users encounter while interacting with a system. These issues can range from navigation difficulties to confusing interface elements.
- **Improving Design:** By identifying usability issues early in the design process or during development, usability testing enables designers to make informed improvements that enhance the overall user experience.
- **Validating Design Choices:** Usability testing validates design decisions by assessing whether the interface supports users in accomplishing tasks smoothly and intuitively.

Key Components of Usability Testing:

- **Participants:** Test participants are recruited based on the target user group of the system. They should represent the diversity of users who will interact with the interface.
- **Tasks:** Usability tests involve defining specific tasks that participants need to complete using the system. These tasks are typically scenarios that users might encounter in real-world use.
- **Observation and Data Collection:** During testing, observers (such as usability experts or designers) watch participants as they perform tasks. They collect data on task completion times, error rates, and participant feedback through methods like think-aloud protocols or post-task interviews.
- **Analysis:** The collected data is analyzed to identify patterns of usability issues and areas where the interface performs well. This analysis informs design changes and improvements.

Heuristic Evaluation

- Heuristic evaluation is a usability inspection method used in human-computer interaction (HCI) to identify usability problems in a user interface design.
- Heuristic evaluation is a usability inspection technique where evaluators systematically examine a user interface design against a set of established usability principles or heuristics.
- These heuristics are broad usability guidelines that help identify potential usability issues without involving actual users in the evaluation process.

Principles of Heuristic Evaluation:

- **Visibility of system status:** Users should be informed about what is happening through appropriate feedback within a reasonable amount of time.
- **Match between system and the real world:** The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms.
- **User control and freedom:** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue.
- **Consistency and standards:** Users should not have to wonder whether different words, situations, or actions mean the same thing.

Components of Heuristic Evaluation:

- **Heuristics:** Heuristics are general rules or guidelines derived from empirical knowledge about usability principles. They are not strict usability rules but rather broadly applicable guidelines that reflect common usability issues.
- **Examples of Heuristics:** Some commonly used heuristics include Nielsen's 10 usability heuristics, which include principles such as visibility of system status, match between system and the real world, and consistency.
- **Evaluators:** Typically, heuristic evaluations are conducted by usability experts, HCI specialists, or experienced designers who are familiar with usability principles and have a good understanding of user interface design.

Task Analysis

- Task analysis involves studying how users perform specific tasks within a system.
- It focuses on identifying the sequence of actions, decision points, cognitive processes, and interactions users engage in to accomplish their goals.

Methods of Task Analysis

- **Observational Studies:** Researchers observe users performing tasks in real or simulated environments. This method provides insights into actual user behaviors, difficulties encountered, and workarounds used.
- **Cognitive Task Analysis:** This approach delves deeper into users' thought processes, decision-making strategies, mental models, and reasoning behind their actions during task performance.
- **Hierarchical Task Analysis (HTA):** HTA breaks down tasks into hierarchical structures, detailing sub-tasks, operations, and actions required at each level. It helps in understanding the complexity of tasks and dependencies between different elements.
- **Critical Incident Technique:** Focuses on critical moments during task execution where errors or difficulties occur, aiming to identify usability issues that impact user performance.
- **Cognitive Walkthrough:** Evaluators simulate user interactions step-by-step from a user's perspective, predicting how users might perform tasks and identifying potential usability problems based on the system's design.

Benefits of Task Analysis in HCI Evaluation:

- **Identifying Usability Issues:** Task analysis helps uncover usability problems such as navigation issues, cognitive overload, unclear instructions, and mismatches between user expectations and system behavior.
- **Improving User Experience:** By understanding how users interact with the system, designers can make informed decisions to streamline workflows, simplify interactions, and enhance overall user experience.
- **Design Validation:** It provides empirical data to validate design decisions, ensuring that user interfaces meet user needs and support efficient task completion.
- **Iterative Design Improvement:** Task analysis supports iterative design processes by continuously evaluating and refining user interfaces based on real-world usage patterns and feedback.

Performance Measures

- Performance measures in the context of human-computer interaction (HCI) assess various aspects of how well users interact with a system.
- These measures help evaluate the effectiveness, efficiency, and satisfaction of user interactions. \
- When evaluating HCI, it's essential to consider a combination of different performance measures to comprehensively assess how well the system supports users in achieving their goals.
- This holistic approach ensures that designers can identify strengths and weaknesses in the user interface, leading to iterative improvements that enhance usability and user experience.

Effectiveness Measures:

- **Task Success Rate:** The percentage of tasks or actions completed successfully by users without errors or assistance.
- **Error Rate:** The frequency and severity of errors users encounter while performing tasks.
- **Accuracy:** The precision with which users perform tasks, often measured by the correctness of inputs or outputs.

Efficiency Measures:

- **Task Completion Time:** The average time taken by users to complete specific tasks or actions.
- **Speed of Performance:** How quickly users navigate through the interface, execute commands, or process information.
- **Workload:** Assessing the cognitive or physical effort required by users to accomplish tasks, often measured through subjective ratings or physiological indicators (e.g., heart rate variability).

Satisfaction Measures:

- **User Satisfaction:** Subjective assessments of users' overall satisfaction with the system, often gathered through surveys, questionnaires, or interviews.
- **Perceived Ease of Use:** Users' perceptions of how easy it is to learn and use the system.
- **User Preferences:** Preferences expressed by users regarding interface design, features, or interaction methods.

Other Performance Measures:

- **Learnability:** The speed and ease with which new users can learn to operate the system.
- **Retention:** The ability of users to retain knowledge of how to use the system over time.
- **Engagement:** Measures of user engagement, such as interaction frequency, duration, or depth of interaction.

Selection of Performance Measures:

- **Contextual Relevance:** Choose measures that align with the specific goals, tasks, and users of the system being evaluated.
- **Quantitative and Qualitative Data:** Use a mix of quantitative metrics (e.g., success rates, time measures) and qualitative insights (e.g., user feedback, preferences) to gain a comprehensive understanding of HCI performance.
- **Iterative Evaluation:** Continuously assess and refine performance measures based on user feedback and evolving system requirements.

Eye Tracking

- Eye tracking is a valuable technique used in human-computer interaction (HCI) research to understand and evaluate how users visually interact with interfaces.
- Here's how eye tracking can be applied for HCI evaluation:
- Eye tracking provides real-time data on where and for how long users look at different elements on the screen.
- This information helps HCI researchers understand:
 - **Visual Attention Patterns:**
 - Usability Evaluation
 - User Experience Insights:
 - Improving Design and Interaction:

Understanding Visual Attention

- Eye tracking provides real-time data on where and for how long users look at different elements on the screen. This information helps HCI researchers understand:
- **Visual Attention Patterns:** Which areas of the interface attract the most attention and which ones are ignored or overlooked.
- **Gaze Sequences:** The order in which users look at interface elements, indicating how they process information and navigate through tasks.
- **Fixation Duration:** How long users dwell on specific elements, indicating cognitive processing and decision-making.

Usability Evaluation:

- Eye tracking can reveal usability issues that might not be detected through traditional methods:
 - **Navigation and Information Processing:** It helps assess how easily users find and process information, revealing potential navigation difficulties or information overload.
 - **Visual Clutter:** Identifies areas of the interface that distract users or cause confusion due to cluttered design.
 - **Attention Allocation:** Determines whether critical elements (e.g., navigation menus, call-to-action buttons) receive adequate visual attention.

User Experience Insights:

Eye tracking contributes to understanding the user experience (UX) by:

- **Emotional Engagement:** Identifying emotional responses through pupil dilation or gaze patterns, which can indicate frustration, interest, or confusion.
- **Visual Appeal:** Assessing the attractiveness and visual appeal of interface designs based on gaze patterns and fixation durations.
- **Learning Curve:** Evaluating how quickly users learn to navigate and interact with the interface based on gaze behaviour over repeated sessions.

Improving Design and Interaction:

- By integrating eye tracking into HCI evaluation, designers can:
- **Iteratively Improve Interfaces:** Use eye tracking data to iteratively refine interface designs, making them more intuitive and user-friendly.
- **Validate Design Decisions:** Validate design decisions based on empirical data rather than subjective opinions alone.
- **Optimize User Interfaces:** Optimize layout, hierarchy, and visual elements to enhance user engagement, efficiency, and satisfaction.

Questionnaire and Survey

- Designing a questionnaire or survey for evaluating human-computer interaction (HCI) involves gathering valuable feedback from users regarding their experiences, perceptions, and satisfaction with the system.

Steps to Design a Questionnaire or Survey for HCI:

1. Define Objectives:

- Determine the specific aspects of HCI you want to evaluate (e.g., usability, user experience, task performance).
- Clarify the goals and research questions you aim to address through the questionnaire/survey.

2. Select Survey Format:

- Decide on the format: online survey, paper-based questionnaire, or in-person interview.
- Choose the appropriate platform or tool for distributing and collecting responses (e.g., Google Forms, SurveyMonkey).

3. Structure and Content:

- **Introduction:** Provide a brief introduction explaining the purpose of the survey, confidentiality assurances, and instructions for completing the survey.
- **Demographic Questions:** Collect basic demographic information (e.g., age, gender, occupation) to understand user characteristics.
- **Main Questions:** Include questions related to the specific aspects of HCI you identified in Step 1. Examples of questions include:
 - **Usability:** "How easy was it to navigate through the system?"
 - **Task Performance:** "Did you encounter any difficulties while completing tasks X and Y?"
 - **User Experience:** "How satisfied are you with the overall design of the system?"
- **Rating Scales:** Use Likert scales (e.g., strongly agree to strongly disagree) or numerical scales to measure attitudes, satisfaction levels, or frequency of behaviors.
- **Open-Ended Questions:** Include opportunities for users to provide qualitative feedback and suggestions. For example:
 - "What aspects of the system do you find most frustrating?"
 - "Do you have any suggestions for improving the system?"
- **Order and Flow:** Organize questions logically, starting with general and easy-to-answer questions before moving to more specific or complex ones.

4. Pilot Testing:

- Conduct a pilot test with a small group of users to identify any ambiguities, confusing questions, or technical issues with the survey instrument.
- Revise and refine the questionnaire/survey based on feedback received during pilot testing.

5. Distribution and Data Collection:

- Distribute the finalized questionnaire/survey to your target audience.
- Ensure clear instructions for completing the survey and provide multiple channels for participants to access and submit their responses (if applicable).

6. Data Analysis:

- Once data collection is complete, analyze the responses quantitatively (e.g., calculating averages, percentages) and qualitatively (thematic analysis of open-ended responses).
- Identify trends, patterns, and areas for improvement based on the survey results.

7. Reporting and Actionable Insights:

- Prepare a report summarizing key findings, including strengths, weaknesses, and recommendations for enhancing HCI.
- Use insights from the survey to inform design decisions and prioritize improvements to the system.

Human Computer Interaction Model and Frameworks

- In human-computer interaction (HCI), various models have been proposed to describe how users interact with computers and other digital devices.
- These models help designers and researchers understand user behavior, improve interface design, and predict user actions.

Types of Model

- Here are some prominent models of interaction in HCI:
 - Human Processor Model
 - GOMS Model
 - KLM Model
 - The Information Processing Cycle
 - Norman's Model of Action
 - Cooperative Model
 - Ucer Cenric design model

key reasons why models and frameworks are essential in HCI

- **Understanding User Behavior:** Models help in understanding how users interact with technology. They provide a structured way to analyze and predict user behavior, which is crucial for designing interfaces that are intuitive and effective.
- **Design Guidance:** Frameworks offer guidelines and principles for designing user interfaces. They provide a structured approach to ensure that design decisions are based on established best practices and theories, rather than intuition alone. This helps in creating interfaces that are usable and meet user needs.
- **Evaluation and Testing:** Models provide a basis for evaluating interactive systems. By using frameworks like ISO 9241-210 or the Cognitive Dimensions Framework, designers can assess usability factors such as effectiveness, efficiency, and satisfaction. This evaluation ensures that systems are evaluated objectively and comprehensively.

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- **Communication and Collaboration:** Models and frameworks provide a common language for designers, developers, and researchers. They facilitate communication by defining terms and concepts, which is crucial for effective collaboration in multidisciplinary teams working on HCI projects.
- **Iterative Improvement:** HCI models support an iterative design process. They encourage designers to prototype, test, and refine designs based on user feedback and empirical data. This iterative approach leads to continuous improvement and innovation in interactive systems.

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- **Education and Training:** Models and frameworks are valuable educational tools. They help in teaching HCI principles and theories to students and practitioners, enabling them to apply theoretical knowledge to real-world design challenges.
- **Cross-Disciplinary Insights:** HCI involves insights from psychology, cognitive science, design, and computer science. Models and frameworks integrate knowledge from these disciplines, providing a holistic view of human-computer interaction and fostering cross-disciplinary research.

GOMS

- The GOMS model (Goals, Operators, Methods, and Selection rules) is a prominent framework in Human-Computer Interaction (HCI) for predicting and analyzing the time it takes to complete tasks.
- It was developed by Card, Moran, and Newell in the early 1980s.

Components of the GOMS model:

- **Goals:** Goals are the overall objectives that a user wants to achieve. In the context of the GOMS model, tasks are decomposed into sub-goals that users need to accomplish to achieve their main objective.
- **Operators:** Operators are the basic cognitive and motor actions that users perform to achieve goals. These actions can include keystrokes, mouse clicks, and mental operations such as decision-making or remembering information.
- **Methods:** Methods are sequences of operators that users employ to accomplish a sub-goal. A method specifies the order in which operators are executed to achieve a particular outcome.
- **Selection rules:** Selection rules are decision points that specify which methods to use under different conditions. They help determine the sequence of actions based on user inputs, system responses, or environmental factors.

HOW GOMS works

- Here's a step-by-step explanation of how the GOMS model operates:
- **Goals Identification:**
 - **Goals** in the GOMS model represent the overall objectives that users aim to achieve when interacting with a system. These goals are defined based on the tasks users need to perform.
- **Operators Definition:**
 - **Operators** are the basic actions or steps that users perform to achieve a goal. They can include physical actions like keystrokes or mouse clicks, cognitive actions like decision-making or information retrieval, and perceptual actions like observing information on the screen.
- **Methods Construction:**
 - **Methods** are sequences of operators that users employ to achieve sub-goals within a task. Each method outlines a specific sequence of actions necessary to accomplish a particular objective.

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- **Selection Rules Application:**

- **Selection rules** determine which methods to use based on different conditions or contexts. They specify decision points within the task flow, guiding the selection of methods according to user inputs, system responses, or environmental factors.

- **Predicting Task Execution Time:**

- The GOMS model, particularly through its extension known as the Keystroke-Level Model (KLM), predicts the time it takes for users to execute tasks by calculating the cumulative time required for each operator involved. KLM assigns fixed times to standard operators (e.g., pressing a key, moving the mouse, perceiving visual information), allowing for a quantitative estimation of task completion time.

Use Case scenarios of GOMS

Case1:

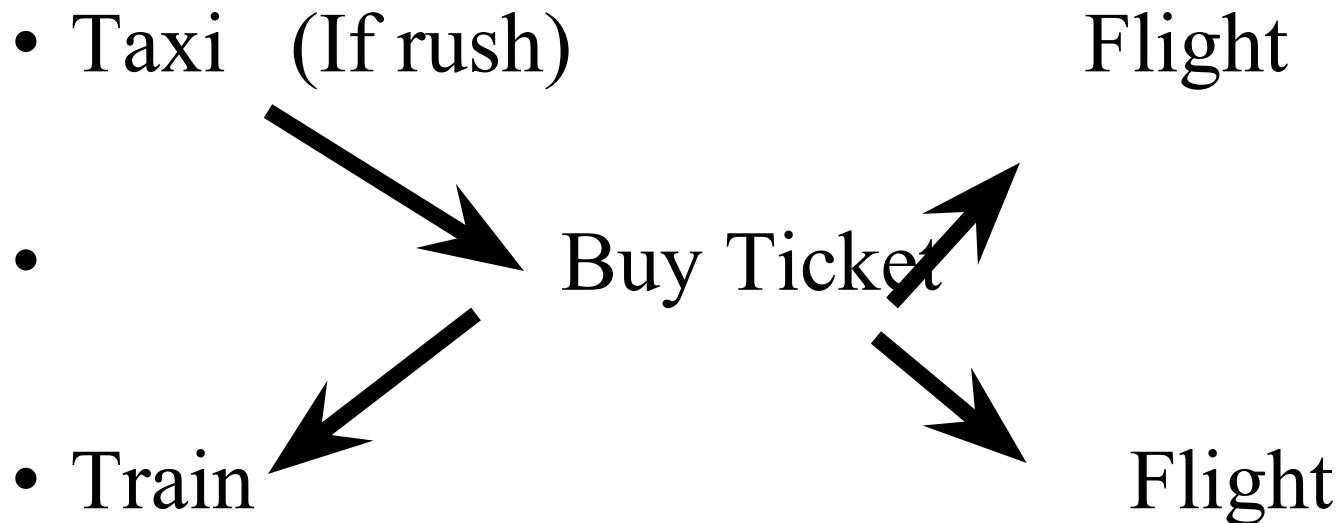
- Goal : Objective is traveling from jaipur to Newyork
- Method: Sequence of Subgoals to achive the main goal
Taxi □ Buy Ticket □ Flight Area
- Operators: One action one can take to achieve a goal or subgoals.
- Taxi Subgoal:
 - Order a cab □ Drive a cab □ Pay for cab
- Buy Ticket Subgoal:
 - Walk □ Walk for Line □ Pay for ticket
- Flight area Subgoal:
 - Wait for flight □ Wait in line □ Board to plane

Case 2

- Goal: Report a product issue
- Operators: Pickup the phone, dial the number
- Open a form in computer, write a letter
- Method : report a problem via phone, Via online form
- Selection: Calling is time consuming, but sometimes fastest way to reach resolution

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- Selection: Deciding factors on how to choose different methods based on certain situations.



Case3:

- Goal: Shutdown the system
- Operator: Click mouse, Press button
- Method: Press Alt+f4, press power button, shutdown
- Selection: Select the method based on user behavior

Key Concepts and Applications of the GOMS Model

- **Keystroke-Level Model (KLM):** A specific application of the GOMS model, the Keystroke-Level Model (KLM) predicts the time it takes to execute each operator (keystroke, mouse click, etc.). It provides a quantitative estimate of task completion time based on the sequence of operators involved.
- **Task Analysis:** GOMS facilitates task analysis by breaking down complex tasks into simpler components (goals, operators, methods). This decomposition helps in understanding how users interact with a system and identifying potential bottlenecks or inefficiencies.
- **Design Evaluation:** Designers use the GOMS model to evaluate interface designs. By simulating user interactions and calculating task completion times, designers can identify interface elements that may hinder efficiency or usability and make informed design decisions.
- **Training and Skill Acquisition:** GOMS can also be used to analyze skill acquisition and training needs. By understanding the sequence of operators and methods required to perform tasks, trainers can develop targeted training programs to improve user proficiency.

Key Stroke Level Model

- keystroke model refers to understanding and utilizing the patterns and characteristics of how users interact with keyboards as an input method.
- The Keystroke-Level Model (KLM) is a predictive model used in human-computer interaction (HCI) to estimate the time it takes for a user to perform a task on a computer interface.
- It's particularly useful for designing and evaluating interfaces based on the cognitive and motor processes involved in completing tasks.

KLM Operators

- In the Keystroke-Level Model (KLM), operators are standardized time values assigned to common cognitive and motor actions that users perform when interacting with a computer interface. These operators are used to estimate the time it takes for a user to complete a task based on the sequence of actions involved. Here are the primary KLM operators:
- **Keystroke (K):**
 - **Time:** Typically 0.2 seconds per keystroke.
 - **Description:** The time it takes for a user to press a key on the keyboard.
- **Pointing (P):**
 - **Time:** Typically 1.1 seconds.
 - **Description:** The time it takes for a user to move the cursor (e.g., with a mouse or trackpad) to a target area on the screen.
- **Button Press (B):**
 - **Time:** Typically 0.1 seconds.
 - **Description:** The time it takes for a user to press a mouse button or similar input device button.
- **Mental Preparation (H):**
 - **Time:** Typically 1.35 seconds.
 - **Description:** The time it takes for a user to prepare mentally before starting a task. This includes interpreting instructions, recalling information, or planning the next action.
- **Drawing (D):**
 - **Time:** Typically 9.0 seconds.
 - **Description:** The time it takes for a user to sketch or draw something on the screen using a pointing device.

- Time are empirically determined
- Time to execute: $TK+TP+TH+TD+TM+TR$

Operators	Remarks	Time (in sec)
K	Press Key Good typist (90 wpm) Poor Typist(40 wpm) Non typist	0.12 s 0.28 sec 1.2 sec
S	Mouse button press down or up click	0.1sec 0.2 Sec
P	Point with mouse flitt's law	1. $\log_2$ (
H	Howm hands to and from keyboard	0.4 sec
D	Drawing domain dependent	
M	Menta preparation	1.35 sec
R	Response from the system	

Use Case for KLM

- **Use Case:** Evaluating Menu Selection in Software Interface Design
- **Scenario:** Imagine a software developer is designing a new graphical user interface (GUI) for a productivity application. One critical element of the interface is the main menu bar, where users navigate to access various functions and commands.

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- **Objective:** The developer wants to evaluate the efficiency of menu selection in the new interface design using the Keystroke-Level Model (KLM).
- **Steps and Analysis:**
 - **Task Definition:** Define the task to be evaluated using the KLM. In this case, it could be selecting a specific command from the main menu using the mouse.
 - **Breakdown into Subtasks:** Divide the task into its constituent subtasks based on the KLM operators:
 - **Pointing:** Moving the mouse cursor to the menu.
 - **Button Press:** Clicking the mouse button to activate the menu.
 - **Keystroke:** Typing the first letter of the command to select it quickly.

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- **Estimating Time with KLM:** Assign time values to each subtask based on the KLM standard time values (in seconds):
 - **Pointing (P):** Typically 1.1 seconds.
 - **Button Press (B):** Usually 0.1 seconds.
 - **Keystroke (K):** Estimated at 0.2 seconds per keystroke.
- For example, if selecting a command involves pointing to the menu (P), clicking (B), and typing the first letter (K), the estimated time TTT using KLM would be:

$$T=P+B+K$$

- If the menu is complex and requires scrolling or additional clicks, these can be added as separate tasks or refinements within the model.

Cont..

- **Evaluation:** Calculate the total estimated time TTT for completing the task and compare it against usability benchmarks or previous interface designs. This evaluation helps in:
 - Identifying potential usability issues (e.g., excessive menu navigation time).
 - Iterating on the interface design to streamline menu access and improve user efficiency.
 - Making informed decisions about interface elements such as menu layout, command organization, and accessibility.

Benefits

- **Quantitative Assessment:** Provides a structured method to estimate and compare task completion times across different interface designs or design iterations.
- **Early Design Feedback:** Allows designers to predict and address potential usability issues early in the design process.
- **Optimization:** Facilitates interface optimization by focusing on reducing task completion times based on cognitive and motor processes.

Applications of KLM

- **Behavioral Analysis:** Keystroke dynamics can be analyzed to understand user behavior and interaction patterns with keyboards. This analysis can provide insights into:
 - Typing speed and accuracy.
 - Error patterns (e.g., common mistyped sequences).
 - How users adapt to different keyboard layouts or input methods.
- **User Profiling:** By analyzing keystroke patterns over time, HCI researchers and designers can build user profiles that help personalize user experiences:
 - Adaptive interfaces that adjust based on typing speed or accuracy.
 - Predictive text or autocomplete features tailored to individual typing patterns.
- **Accessibility:** Keystroke modeling plays a role in designing accessible interfaces:
 - Understanding how different users with varying physical abilities interact with keyboards can inform the design of inclusive interfaces.
 - Adaptive keyboards or input methods that adjust sensitivity or layout based on user behavior.

- **Error Detection and Correction:** Keystroke models can aid in error detection and correction strategies:
 - Identifying frequent mistypes or errors can inform the design of predictive text or autocorrect algorithms.
 - Providing feedback to users on typing accuracy to improve overall user experience.
- **Security:** In addition to authentication purposes mentioned earlier, keystroke dynamics can contribute to securing interactions and data entry:
 - Detecting anomalies in typing behavior that may indicate unauthorized access attempts or compromised user accounts.
 - Enhancing security through continuous monitoring of typing patterns as a secondary authentication factor.
- **Training and Assistance:** HCI can leverage keystroke models to provide training and assistance to users:
 - Teaching touch typing techniques or improving typing efficiency based on individual user behavior.
 - Providing feedback on typing habits to help users develop better ergonomic practices.